Name: Student ID:

Here is a contingency table of college students with their **Favorite Color** (Red or Blue) down the columns and their **School** (Berkeley or Stanford) across the rows.

| | Red | Blue |
|---|---|---|
| Berkeley | 10 | 90 |
| Stanford | 60 | 40 |

1. a) Find the proportion of all students who attend Berkeley and b) what type of proportion is this?

2. a) Find the proportion of all students who attend Berkeley *and* like red best and b) what type of proportion is this?

3. a) *Of the students who attend Berkeley*, find the proportion that like red best and b) what type of proportion is this?

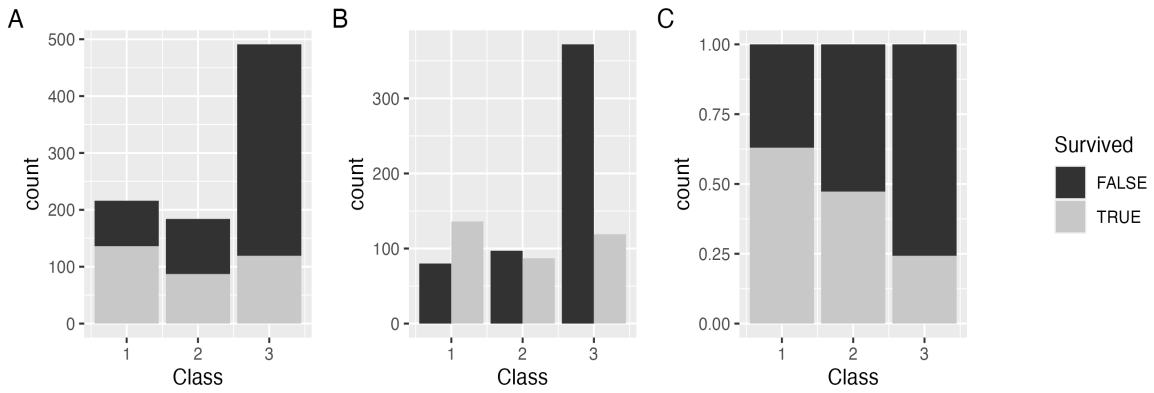
RMS Titanic was an infamous British passenger. It sank in the North Atlantic Ocean on 15 April 1912 after striking an iceberg on her way to New York City. The wreck the deadliest sinking of a single ship at the time, with almost 70% of the passengers and crew dying.

Below are the first five rows of for a data frame called `titanic`, which includes information on some of the passengers.

| Name | Sex | Age | Fare | Class | Survived |
|---|---|---|---|---|---|
| Braund, Mr. Owen Harris | male | 22 | 7.25 | 3 | FALSE |
| Cumings, Mrs. John Bradley (Florence Briggs Thayer) | female | 38 | 71.28 | 1 | TRUE |
| Heikkinen, Miss. Laina | female | 26 | 7.92 | 3 | TRUE |
| Futrelle, Mrs. Jacques Heath (Lily May Peel) | female | 35 | 53.10 | 1 | TRUE |
| Allen, Mr. William Henry | male | 35 | 8.05 | 3 | FALSE |

4. What is the unit of observation for `titanic`?

The three following graphs were generated using `titanic`.




5. Which graph is **best** used to answer the following questions? *Write the letter associated with the graph to the right of each question*

- *How many first class passengers survived the Titanic's sinking?* _____
- *Is there an association between a passenger's class and their survival?* _____
- *What is the mode of the `Class` variable?* _____

6. Roughly how many passengers are in the `titanic` data frame, based on the graphs?

7. The `promote` data frame can be found in the `stat20data` R package. Using `promote`, write `ggplot2` code to make a stacked, normalized bar chart having identified gender on the x axis and the bars filled in by promotion decision. Once your code is working, copy your code into the box to the right. *Make sure you load any necessary packages!*

8. Write a one-sentence claim about the association between identified gender and promotion decision.

9. Let's practice using functions and help pages. When a dataset has thousands of rows, viewing a random sample can make it easier to preview. For example, `slice_sample` is a useful function from `tidyverse` to pull just a few rows from a dataframe. One way to use it is like so: `slice_sample(penguins, n = 5)` will return five random rows from `penguins`. Copy the code into R studio and run it to confirm.

```
library(tidyverse)
library(stat20data)
slice_sample(penguins, n = 5)
```

Now change the last line so it returns a random **2% of all rows**, without first finding the number of rows. Some hints:

- Open the help page by running `?slice_sample` in a new cell. Look at the page that opens in the bottom-right, and find the argument for sampling a proportion rather than a fixed number.
- Replace `n = 5` with that argument and the appropriate value for 2%.
- Once it works (you should get 6 rows), write your revised `slice_sample()` line below.